Customer Shopping Behaviour Analysis – Project Report

10 November 2026

Executive Summary

The 3,900 customers in this dataset generated $233,081 in revenue at an average of $59.76 per purchase, and that average barely moves across any customer segment we tested. Growth will therefore come from more customers, more repeat purchases and smarter promotions, not from targeting a high-spending group.

This project cleaned the raw data in Python, loaded it into PostgreSQL, answered ten business questions in SQL, and visualised the results in a Power BI dashboard. The main findings:

- **Spend is flat across segments.** Average purchase is $59.49 for subscribers vs $59.87 for non-subscribers, $59.28 with a discount vs $60.13 without, and $59.07–$60.45 across the four age groups.
- **Men drive volume, not value.** Men account for 67.7% of revenue ($157,890) because they are 68% of customers; per-purchase spend is almost identical ($59.54 men, $60.25 women).
- **The base is already loyal.** 3,116 customers (80%) have more than 10 previous purchases, yet only 27% of customers subscribe, so loyalty is not being converted into subscriptions.
- **Discounts are heavily used but do not lift basket size.** 43% of purchases carry a discount, and every subscriber receives one.
- **Clothing is the core category**, with 44.7% of revenue ($104,264), followed by Accessories (31.8%).

Business Problem and Objectives

A leading retail company wants to understand its customers' shopping behaviour to improve sales, customer satisfaction and long-term loyalty. Management has seen purchasing patterns shift across demographics and product categories, and wants to know which factors (discounts, reviews, seasons, payment preferences) drive purchase decisions and repeat buying.

The overarching question for this project:

> How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimise marketing and product strategies?

To answer it, the project had four objectives:

1. Clean and transform the raw customer dataset in Python so it is ready for analysis.
2. Store the data in a structured PostgreSQL table and use SQL to analyse customer segments, loyalty and purchase drivers.
3. Build an interactive Power BI dashboard that lets stakeholders explore key patterns.

4. Summarise findings and recommendations in this report.

Dataset Description

The dataset (`customer_shopping_behavior.csv`) has 3,900 rows and 18 columns, with one row per customer and one purchase recorded for each. Customer IDs run from 1 to 3,900, and customers are aged 18 to 70.

Field group	Columns	Notes
Customer	customer_id, age, gender, location	68% male, 32% female; 50 US states
Product	item_purchased, category, size, color, season	25 items in 4 categories (Clothing, Accessories, Footwear, Outerwear)
Transaction	purchase_amount (USD), payment_method, shipping_type	Purchase amount ranges $20–$100; 6 payment methods, 6 shipping types
Promotion	discount_applied, promo_code_used, subscription_status	Discount and promo code are identical in every row
Behaviour	review_rating, previous_purchases, frequency_of_purchases	Rating 2.5–5.0; previous purchases 1–50; 7 frequency labels

The only missing values were 37 review ratings (0.9% of rows). The dataset has no date or sales-channel field, which limits some of the questions in the business problem (see Limitations).

Tools and Methodology

The project is a three-stage pipeline, with each tool doing the job it is best at.

1. **Python (pandas, SQLAlchemy):** load the CSV, clean it, engineer new columns, then write the result to PostgreSQL.
2. **PostgreSQL:** hold the cleaned data in a `customer` table and answer ten business questions with SQL (aggregations, subqueries, CTEs, CASE segmentation and window functions).
3. **Power BI:** connect to the same table and build an interactive dashboard with KPI cards, charts and slicers.

Using one cleaned table as the single source means the SQL results and the dashboard always agree.

Data Preparation and Feature Engineering

Five cleaning and modelling steps were applied in Python before loading the data into PostgreSQL.

Step	What was done	Why
Missing ratings	Filled 37 missing `review_rating` values with the median rating of the same category	Category median keeps each category's rating profile intact and is not skewed by outliers
Column names	Converted to lowercase snake_case; `purchase_amount_(usd)` became `purchase_amount`	Makes columns safe to use in SQL without quoting
Age groups	Split `age` into quartiles with	Gives four equal-sized groups for

Step	What was done	Why
	`pd.qcut`: Young Adult (18–31), Adult (32–44), Middle-aged (45–57), Senior (58–70)	demographic comparison
Purchase frequency	Mapped `frequency_of_purchases` text to days (Weekly 7, Fortnightly 14, Monthly 30, Quarterly 90, Annually 365)	Turns a label into a number that can be averaged or compared
Redundant column	Dropped `promo_code_used` after confirming it equals `discount_applied` in all 3,900 rows	Removes duplicate information

The cleaned table was then written to the `customer` table in a PostgreSQL database through SQLAlchemy.

The frequency labels contain overlapping categories (Fortnightly and Bi-Weekly both map to 14 days; Quarterly and Every 3 Months both map to 90 days), so the source data likely uses two labels for the same behaviour.

SQL Analysis: Ten Business Questions

The ten queries show that customers differ little in how much they spend, and that the real differences lie in who the customers are and how they are promoted to. All figures come from the 3,900-row `customer` table.

#	Business question	Result	What it means
1	Revenue by gender	Male $157,890 (67.7%); Female $75,191 (32.3%)	Men bring more revenue because there are twice as many male customers; average spend is almost equal ($59.54 vs $60.25)
2	Customers who used a discount and still spent above the $59.76 average	839 of the 1,677 discounted purchases (50%)	Half of discounted buyers would likely have spent above average anyway, so discounts may be going to customers who did not need them
3	Top 5 products by average review rating	Gloves 3.86, Sandals 3.84, Boots 3.82, Hat 3.80, Skirt 3.78	Ratings sit in a narrow 3.7–3.9 band; Skirt, T-shirt and Handbag tie at 3.78 after rounding
4	Average purchase: Standard vs Express shipping	Express $60.48; Standard $58.46	Express buyers spend about $2 more per order, a small gap
5	Do subscribers spend more?	Subscribers: 1,053 customers, $59.49 average, $62,645 revenue. Non-subscribers: 2,847 customers, $59.87 average, $170,436 revenue	Subscribers do not spend more per purchase; they are 27% of customers and 27% of revenue
6	Top 5 products by share of purchases with a discount	Hat 50.0%, Sneakers 49.7%, Coat 49.1%, Sweater 48.2%, Pants 47.4%	These products depend most on discounts; the overall discount rate is 43%

#	Business question	Result	What it means
7	Customer segments by previous purchases	New (1 purchase): 83; Returning (2–10): 701; Loyal (11+): 3,116	80% of customers are Loyal, so the segment thresholds do not separate customers well
8	Top 3 products per category	Clothing: Blouse 171, Pants 171, Shirt 169. Accessories: Jewelry 171, Belt 161, Sunglasses 161. Footwear: Sandals 160, Shoes 150, Sneakers 145. Outerwear: Jacket 163, Coat 161	Order counts are close within each category, so no single product dominates; Outerwear has only two items
9	Are repeat buyers (more than 5 previous purchases) more likely to subscribe?	958 of 3,476 repeat buyers subscribe (27.6%) vs 27.0% of all customers	Repeat buying has no visible link to subscribing
10	Revenue by age group	Young Adult (18–31) $62,143 (26.7%); Middle-aged (45–57) $59,197 (25.4%); Adult (32–44) $55,978 (24.0%); Senior (58–70) $55,763 (23.9%)	Revenue is spread almost evenly; no age group is a clear priority

Two supporting results add context. By category, Clothing earns $104,264 (44.7%), Accessories $74,200 (31.8%), Footwear $36,093 (15.5%) and Outerwear $18,524 (7.9%). By season, Fall has the highest average purchase ($61.56) and Summer the lowest ($58.41).

Power BI Dashboard

The Customer Behavior Dashboard is a single interactive page connected to the PostgreSQL `customer` table, built so a manager can answer common questions without writing SQL.

- **KPI cards:** number of customers (3,900), average purchase amount ($59.76) and average review rating (3.75).
- **Subscription mix:** a donut chart of the percentage of customers by subscription status.
- **Revenue views:** revenue by age group and revenue by category.
- **Volume views:** total customers by category and total orders by age group.
- **Slicers:** gender, subscription status, shipping type and category, which filter every chart on the page at once.

The slicers make it easy to test a segment directly, for example subscribers only, or Footwear buyers who chose Express shipping. The dashboard file is `customer_behavior_dashboard.pbix` in the project repository.

Key Insights

Management asked which factors drive purchases and repeat buying. In this dataset, none of the factors tested changes spend per purchase by more than about $3, so the insights concern customer mix and promotion design rather than a single dominant driver.

- **Discounts:** 43% of purchases use a discount, but discounted purchases average $59.28 against $60.13 without one. Discounts are not lifting basket size.

- **Subscriptions:** every subscriber (27% of customers) receives a discount, against 21.9% of non-subscribers, yet subscribers spend no more ($59.49 vs $59.87).
- **Reviews:** average ratings are 3.72–3.79 in every category and 3.7–3.9 for the best products. No category or product stands out as a satisfaction problem or strength.
- **Seasons:** Fall has the highest average purchase ($61.56) and revenue ($60,018). Summer has the lowest average ($58.41) and revenue ($55,777).
- **Payment methods:** all six are used almost equally, from 612 (Bank Transfer) to 677 (PayPal) customers, so no payment option needs priority.
- **Loyalty:** 80% of customers have more than 10 previous purchases, and purchase frequency is spread evenly across all seven labels (539–584 customers each).
- **Customer mix:** men are 68% of customers and Clothing is 44.7% of revenue, so the business is concentrated by who buys and what they buy, not by how much each buys.

Business Recommendations

Because spend per purchase is flat, the recommendations aim to grow the number of customers and purchases, and to stop spending on discounts that do not change behaviour.

1. **Target discounts instead of giving them broadly.** Half of discounted purchases were above the average amount. Run an A/B test that withholds discounts from a sample of customers and compare their spend and repeat rate.
2. **Give subscribers more than a discount.** Subscribers get a discount on every purchase but spend the same as non-subscribers. Test perks such as free shipping, early access or points, and measure whether they raise purchase frequency.
3. **Convert loyal customers into subscribers.** 80% of customers are already repeat buyers, yet only 27% subscribe. Offer subscription to customers with 10 or more previous purchases first.
4. **Grow the female customer base.** Women are 32% of customers but spend as much per purchase as men, so every extra female customer is worth the same revenue. Prioritise acquisition campaigns aimed at them.
5. **Plan promotions around seasons.** Fall has the strongest average spend, so protect margin there. Use promotions in Summer and Spring, when average spend is lowest ($58.41 and $58.74).
6. **Review the Outerwear range.** It has only two products and 7.9% of revenue. Either expand the range or confirm it deserves less shelf and marketing space.
7. **Redefine the loyalty tiers.** With 80% of customers labelled Loyal, the label tells management nothing. Use higher cut-offs (for example 40 or more previous purchases) or percentiles to find the top customers.

Limitations and Conclusion

The analysis answers what the dataset can show, but it cannot answer every part of the business problem.

- **No sales channel field:** the problem mentions online vs offline sales, but the data has no channel column. Shipping type (for example Store Pickup) is only a rough substitute.

- **No dates:** each customer has one purchase and no timestamp, so sales trends over time cannot be measured; season is the only time signal.
- **No cost or margin data:** discounts are judged on spend only, not profit.
- **Discounts and subscriptions overlap:** every subscriber receives a discount, so the effect of the two cannot be separated.
- **Imputed ratings:** 37 missing ratings (0.9%) were filled with category medians, which has a negligible effect on averages.
- **Segment thresholds:** the New, Returning and Loyal cut-offs put 80% of customers in one group.

Conclusion. The company's customers are loyal, evenly spread across ages, payment methods and purchase frequencies, and spend about $60 per purchase regardless of segment. The opportunities are to target discounts, make subscriptions worth more than a price cut, and widen the customer base beyond its male, Clothing-heavy core. Adding date, channel and cost data would let the next round of analysis measure trends and profitability.